

Lecture 5: Space Weather, Galactic Cosmic Rays, and the Atmospheric Spallation Source

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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1 Introduction: The Earth as a Nuclear Target

In our study of nuclear engineering thus far, we have largely focused on **closed systems**: a fuel pellet, a reactor core, or a lead-shielded source. In these environments, we control the flux and the geometry. However, from a geochemical and geophysical perspective, the Earth itself is a giant, unshielded target in a high-energy radiation field.

1.1 The "Open System" Perspective

Most standard textbooks, including Lamarsh, treat the Earth as an isolated system where radionuclides are either "primordial" (left over from the Earth's formation, like ^{238}U) or "anthropogenic" (man-made, like ^{137}Cs).

This perspective ignores a third, vital category: **Cosmogenic Radionuclides**. These are isotopes constantly being "manufactured" in the atmosphere by the interaction of cosmic rays with stable gas molecules. Without this external nuclear "rain," tools like Carbon-14 dating would not exist, as the primordial inventory of Carbon-14 would have decayed away billions of years ago.

1.2 Defining "Space Weather"

In a nuclear context, **Space Weather** is the study of the varying fluxes of particles and electromagnetic radiation originating from the Sun and the Galaxy. For a Nuclear Engineer, Space Weather represents the "Source Term" for the atmospheric reactor.

- **Kinetic Space Weather:** High-velocity particles (protons, alphas) that trigger nuclear transmutations. This is the driver for ^{14}C production.
- **Electromagnetic Space Weather:** X-rays and solar flares that impact the ionosphere and can induce massive currents in the power grid (e.g., the Carrington Event).

1.3 The Atmospheric Shield

We often think of the atmosphere only as something we breathe, but in nuclear terms, it is a **biological shield**.

- The atmosphere has an "areal density" of approximately 1030 g/cm^2 (the mass of the air column above a square centimeter at sea level).
- In terms of radiation protection, this is equivalent to roughly **10 meters (33 feet) of water** or **3 feet of steel**.

1.4 Engineering Context: Why this matters

Understanding the Earth as a target is no longer just for geologists. Modern engineers must account for this "target" behavior in several fields:

1. **Aerospace Engineering:** Pilots and frequent flyers receive a measurable increase in dose because they are "higher up" in the shield.
2. **Microelectronics:** As transistors get smaller, a single secondary neutron from a cosmic ray can flip a bit in a memory chip, causing a "Single Event Upset" (SEU).
3. **Environmental Monitoring:** To accurately measure man-made leaks from a reactor, we must first subtract the "background" created by the atmospheric spallation we will discuss today.

Core Concept: The Carbon-14 clock is not a closed system; it is a **steady-state inventory** where the production rate (P) from space weather exactly balances the decay rate (λN) of the global inventory.

1.5 Reference for Further Reading

- **UNSCEAR (2000) Report:** [Annex B: Exposures from Natural Radiation Sources](#). This document provides a rigorous breakdown of cosmic radiation flux, altitude effects, and the global production rates of cosmogenic radionuclides.

2 The Sun: Our Local Plasma Source

The Sun is the most prominent feature of our local radiation environment, but from a nuclear transmutation perspective, it is surprisingly "quiet." To understand why, we must look at the kinematics of the Solar Wind.

2.1 The Solar Wind: Composition and Velocity

The Solar Wind is a plasma consisting primarily of ionized hydrogen (protons) and electrons, with a small fraction of alpha particles. This plasma is not "radiated" like light; it is physically ejected from the solar corona and flows outward throughout the solar system.

- **Velocity:** Typically ranges from 300 to 800 km/s (often categorized as "slow" or "fast" solar wind).
- **Energy Profile:** While these speeds are immense by terrestrial standards, the *kinetic energy* of a proton moving at 500 km/s is only about **1.3 keV**. Even the most energetic particles in a standard solar stream rarely exceed **10 keV**.

2.2 The Earth's First Line of Defense: The Magnetosphere

Because the Solar Wind consists of charged particles, it is subject to the Lorentz force ($\mathbf{F} = q\mathbf{v} \times \mathbf{B}$). The Earth's intrinsic magnetic field creates a "bubble" known as the **Magnetosphere**.

- **Deflection:** The vast majority of solar protons are deflected around the Earth at the **Bow Shock**, similar to water flowing around the prow of a ship.

- **The Aurora:** Some particles are captured and funneled along magnetic field lines toward the North and South poles. When these keV-range protons collide with Nitrogen and Oxygen in the upper atmosphere, they excite the electrons in those atoms. The resulting "de-excitation" releases the visible light we recognize as the **Aurora**.

2.3 The Nuclear "Dead End"

For a Nuclear Engineer, the most important takeaway is the **Coulomb Barrier**.

- To trigger a nuclear reaction (like spallation or capture), a proton must get close enough to a nucleus for the Strong Force to take over. This requires overcoming the electrostatic repulsion of the target nucleus.
- The Coulomb barrier for a Nitrogen nucleus is approximately **2–3 MeV**.
- **Conclusion:** Solar wind protons (~ 1 keV) are three orders of magnitude too weak to cause nuclear transmutations. They cause *atomic* excitations (light) and *electronic* disruptions, but they cannot create Carbon-14.

Reference for Further Reading

- **NASA Space Physics:** [The Solar Wind Tutorial](#). This provides the specific flux and density data for solar plasma at 1 AU.
- **Kallenrode, M. B. (2004):** *Space Physics: An Introduction to Plasmas and Particles in the Heliosphere and Magnetospheres*. Springer. (A standard textbook reference for the energy-dependent behavior of solar vs. galactic particles).

3 The Carrington Event: EM vs. Kinetic Energy

While we have established that solar particles generally lack the kinetic energy to trigger nuclear transmutations, the Sun is capable of massive electromagnetic (EM) outbursts. In nuclear engineering, we must distinguish between "Hard Radiation" (particles) and "Inductive Effects" (fields).

3.1 Coronal Mass Ejections (CMEs)

A CME is a massive burst of solar wind and magnetic fields rising above the solar corona or being released into space. Unlike the steady-state solar wind, a CME is a coherent "slug" of plasma that carries an immense frozen-in magnetic field.

3.2 The 1859 Carrington Event

The most famous example occurred in September 1859, observed by British astronomer Richard Carrington.

- **The Observation:** A "white-light flare" was seen on the solar disk, followed approximately 17 hours later by a massive geomagnetic storm.
- **The Impact:** The Earth's magnetosphere was severely compressed. Auroras were visible as far south as the Caribbean.

- **Engineering Consequence:** The rapidly changing magnetic field ($\frac{dB}{dt}$) induced massive currents in long-wire telegraph systems. Operators reported sparks leaping from their equipment, and in some cases, the lines continued to send signals even after the batteries were disconnected.

3.3 The Engineering Lesson: Ionizing vs. Inductive

For the Nuclear Engineer, this distinction is critical for system design:

1. **Ionizing Radiation:** Requires high kinetic energy (MeV/GeV) to penetrate shielding or flip bits in microchips. This is largely a **Galactic** problem.
2. **Geomagnetically Induced Currents (GIC):** Requires a massive, moving magnetic field. This is a **Solar** problem. A modern "Carrington-level" event would likely destroy high-voltage power transformers across the globe, not through radiation damage, but through thermal failure due to induced DC currents saturating the magnetic cores.

Reference for Further Reading

- **Space.com:** [The Carrington Event: History's greatest solar storm](#). A historical and technical review of the Carrington observations with lots of additional references.

4 Crossing the Border: The Heliopause

To understand where the Carbon-14 "fuel" actually comes from, we have to look far beyond the Earth, past the orbit of Pluto, to the edge of the Sun's physical influence.

4.1 The Heliosphere: Our Magnetic Bubble

The solar wind we discussed in Section 2 doesn't just stop at the planets; it flows outward in all directions, creating a giant "bubble" in the interstellar medium known as the **Heliosphere**.

- Inside this bubble, the environment is dominated by solar plasma and the Sun's magnetic field.
- Outside this bubble lies the "Interstellar Medium" (ISM)—the debris, gas, and high-energy radiation of the Milky Way galaxy.

4.2 The Heliopause: The Pressure Balance

The **Heliopause** is the theoretical boundary where the outward "dynamic pressure" of the solar wind ($P = \frac{1}{2}\rho v^2$) exactly balances the inward pressure of the interstellar gas.

4.3 The Voyager Discovery (The "In-Situ" Experiment)

For decades, our understanding of the Heliopause was purely mathematical. That changed with the Voyager 1 and 2 missions.

- **The Crossing:** In August 2012, Voyager 1 became the first human-made object to cross the Heliopause.

- **The Data:** Scientists observed a dramatic, sudden drop in solar particles and a simultaneous, massive **surge in high-energy Galactic Cosmic Rays (GCRs)**.
- **The "Shield" Confirmed:** This measurement proved that the Sun's magnetic wind acts as a giant "shield" (via magnetic scattering and diffusion), blocking approximately **90%** of the galaxy's lower-energy cosmic rays from ever reaching the inner solar system.

4.4 Implications for the C-14 Clock

This discovery is critical for the "Nuts and Bolts" of Carbon-14 dating:

1. **The Source Baseline:** We now know the "unmodulated" flux of the galaxy.
2. **The Modulation Effect:** Because the solar wind strength varies with the 11-year solar cycle, the "thickness" of this shield changes. This causes the production rate of Carbon-14 on Earth to fluctuate slightly over time—one of the primary reasons we must "calibrate" the Carbon-14 clock against known-age samples like tree rings.

Reference for Further Reading

- **NASA Jet Propulsion Laboratory (JPL):** [Voyager: The Interstellar Mission](#). This is the official mission page detailing the transition into interstellar space and the specific instruments used to measure the cosmic ray surge.
- **Technical summary of recent Voyager results:** [Galactic Cosmic Rays Throughout the Heliosphere and in the Very Local Interstellar Medium](#).

5 The Galactic "Violent" Source (GCRs)

Now that we have moved outside the protective "bubble" of the Sun, we encounter the true source of our atmospheric nuclear reactions: **Galactic Cosmic Rays (GCRs)**.

5.1 What are GCRs?

Despite the name "rays," these are not electromagnetic radiation (like X-rays or Gamma rays). They are high-mass, relativistic charged particles.

- **Composition:** Approximately 89% protons (H^+), 10% alpha particles (He^{++}), and 1% heavier nuclei (up to Iron).
- **The Origin:** Accelerated to near-light speeds by the massive shockwaves of **Supernovae** occurring across our galaxy.

5.2 Energy: The GeV Scale

While we measure Solar Wind in keV, GCRs are measured in **GeV (Giga-electron volts)** and even **TeV**.

- A typical GCR proton has a kinetic energy of 1 GeV to 10 GeV.
- **The Engineering Comparison:** The protons in the Large Hadron Collider (LHC) are accelerated to 6.5 TeV. Nature is hitting the top of our atmosphere with particles of comparable—and sometimes much higher—energies.

5.3 Penetration: Punching through the Magnetosphere

In Section 2, we saw that the Earth’s magnetic field easily deflects the 1 keV solar wind. However, for a 10 GeV GCR proton, the Earth’s field is essentially ”transparent.”

- **Magnetic Rigidity:** This is a measure of a particle’s resistance to being deflected by a magnetic field. Because GCRs have such high momentum, their *gyroradius* is often larger than the Earth’s magnetosphere itself.
- **Impact:** These particles do not get funneled to the poles like the solar wind; they strike the atmosphere at all latitudes (though there is a slight ”latitude effect” where the field is weaker at the poles).

5.4 The Biological/Engineering Shield (Revisited)

Because GCRs are so energetic, they represent the primary radiation hazard for deep-space travel (e.g., a mission to Mars).

- **Secondary Radiation:** When a GCR hits a metal spacecraft wall, it creates a shower of secondary neutrons and pions—ironically making the radiation environment *inside* a poorly designed ship worse than the one outside.
- **Earth’s Advantage:** We are protected not by a magnetic field, but by the **mass** of our atmosphere. The GCRs must pass through $\sim 1000 \text{ g/cm}^2$ of air, which acts as a ”spallation target” that absorbs their energy.

Reference for Further Reading

- **NASA Goddard Space Flight Center:** [Cosmic Rays](#). A clear breakdown of GCR composition and the ”Extra-Solar” origin.
- **The Voyager Interstellar Mission:** [Voyager 1 & 2 Mission Profile](#).

6 The Atmospheric Spallation Source

In a standard nuclear reactor, we rely on **fission** (splitting heavy nuclei with low-energy neutrons). In the upper atmosphere, we see **spallation**: the ”shattering” of light nuclei by ultra-high-energy particles.

6.1 The Physics of Spallation

When a GeV-scale GCR proton strikes a nucleus (primarily ^{14}N or ^{16}O), the wavelength of the projectile is much smaller than the dimensions of the nucleus.

- **The Intranuclear Cascade:** The proton doesn’t ”bounce off”; it plow through, colliding with individual nucleons. This ”cue ball” effect ejects a spray of secondary particles.
- **Evaporation:** The remaining ”heated” nucleus sheds its excess energy by ”boiling off” additional neutrons and protons.

6.2 The Secondary "Debris" Flux

A single primary GCR proton can generate a "shower" of thousands of secondary particles. For our purposes, the most important are:

- **Neutrons (n):** These are the "parent" particles for ^{14}C .
- **Muons ($\mu^\pm$):** Highly penetrating particles that reach the Earth's surface. (Fun fact: about 10,000 muons pass through your body every minute).
- **Pions (π):** Short-lived mesons that decay into the muons and gammas we detect at sea level.

6.3 Engineering Tie-in: The SNS (Oak Ridge)

This is not just "space physics"—it is a major tool in material science.

- At the **Spallation Neutron Source (SNS)** in Oak Ridge, TN, engineers use a linear accelerator to fire 1 GeV protons at a liquid Mercury target.
- **The Advantage:** Unlike a reactor, which provides a steady (DC) flux of thermal neutrons, a spallation source provides an extremely intense, **pulsed** (AC) flux of high-energy neutrons.
- The Earth's upper atmosphere is essentially a "Natural SNS," where the "target" is the Nitrogen gas.

Reference for Further Reading

- **Oak Ridge National Laboratory:** [Virtual Tour](#). A virtual tour of the SNS facility, with links to other resources.
- **Wikipedia:** [Cosmogenic Nuclides](#). A useful discussion of all the nuclides formed from cosmic ray bombardment.

7 The Production of Carbon-14

Once the spallation process in the upper atmosphere has liberated a "sea" of neutrons, the final nuclear synthesis of ^{14}C can begin. This requires two distinct steps: Moderation and Capture.

7.1 Neutron Moderation: Slowing Down

The neutrons produced by spallation are "fast" (carrying MeV of energy). However, Nitrogen-14 has a much higher capture cross-section for "thermal" neutrons (energy ≈ 0.025 eV).

- **The Moderator:** The atmosphere itself (primarily N_2 and O_2 molecules) acts as the moderator.
- **Elastic Scattering:** Through repeated collisions with these nuclei, the neutrons lose kinetic energy until they reach thermal equilibrium with the surrounding gas.

7.2 The (n, p) Reaction

The primary channel for ^{14}C production is an **exothermic (n, p) reaction** with the most abundant isotope in our atmosphere, Nitrogen-14.



- **The Mechanics:** The thermal neutron is absorbed by the ^{14}N nucleus, forming an excited intermediate state which then ejects a proton to reach the relatively stable (but radioactive) ^{14}C state.
- **Energy Balance:** The reaction is exothermic, releasing about 626 keV of energy, which is carried away by the kinetic energy of the ejected proton and the recoiling ^{14}C nucleus.

7.3 Chemical Fate: From Nucleus to CO_2

As a Nuclear Engineer, you must ask: "What happens to the atom after the reaction?"

1. The nascent ^{14}C atom is highly reactive (a "hot atom").
2. It quickly reacts with atmospheric Oxygen to form ^{14}CO (**Carbon Monoxide**) and eventually $^{14}\text{CO}_2$ (**Carbon Dioxide**).
3. This $^{14}\text{CO}_2$ is chemically identical to standard $^{12}\text{CO}_2$, allowing it to enter the global carbon cycle through photosynthesis.

7.4 Other Cosmogenic "Debris"

While ^{14}C is the most famous, other capture and spallation channels produce a suite of radionuclides:

- **Tritium (^3H):** Produced via $^{14}\text{N} + n \longrightarrow {}^{12}\text{C} + {}^3\text{H}$.
- **Beryllium-7 and 10:** Produced by the direct spallation (fragmentation) of Nitrogen and Oxygen.

Reference for Further Reading

- **Wikipedia: [Carbon-14](#).** (A reliable summary of the (n, p) reaction and the discovery of the isotope by Kamen and Ruben in 1940).

8 Space Engineering Implications

The high-energy environment that creates our "Atmospheric Nuclear Reactor" poses significant challenges for modern engineering, particularly as we move beyond the protective "shield" of Earth's atmosphere and magnetosphere.

8.1 Deep Space Travel: The Shielding Paradox

One of the most significant hurdles for a manned mission to Mars is not the Sun, but the Galactic Cosmic Rays (GCRs) we discussed in Section 5.

- **Transient vs. Continuous:** Solar flares (Solar Particle Events) are intense but transient; astronauts can retreat to a "storm shelter" made of hydrogen-rich material (like water or polyethylene) for a few hours.
- **The GCR Constant:** GCRs are a continuous, omnidirectional "background" of GeV-scale particles. Because they are so energetic, they are nearly impossible to stop with conventional shielding.
- **Secondary Spallation:** When a GeV proton strikes a heavy metal shield (like lead or aluminum), it triggers spallation *within the shield itself*, creating a shower of secondary neutrons and X-rays. This can actually increase the biological dose to the crew—a phenomenon known as the "shielding paradox."

8.2 Electronics Hardening: Single Event Upsets (SEUs)

As we shrink the size of transistors on silicon chips to the nanometer scale, the amount of charge required to represent a "1" or a "0" (the critical charge) becomes tiny.

- **Bit Flipping:** When a secondary neutron or a heavy cosmic ray ion passes through a semiconductor, it leaves a trail of ionization. This can deposit enough charge to flip a bit in memory or a logic gate.
- **SEUs:** This is known as a **Single Event Upset**. While usually not destructive to the hardware, it causes software "glitches" that can be catastrophic for flight control systems.
- **Rad-Hardening:** Spacecraft engineers must use "Rad-Hard" components, which often involve redundant logic (e.g., Triple Modular Redundancy, where three chips "vote" on every calculation) or specialized manufacturing processes like Silicon-on-Insulator (SOI).

Reference for Further Reading

- **NASA Goddard:** [Space Radiation is Risky Business](#). A high-level overview of the biological and engineering risks of GCRs.
- **IEEE Spectrum:** [How to Kill a Supercomputer](#). A fascinating look at how cosmic rays cause errors even in terrestrial computers at high altitudes or in large data centers.

9 Conclusion: From Cosmic Rays to Global Tracers

We have seen that the Earth is not a closed system, but a target in a high-energy galactic laboratory. The constant "rain" of cosmic rays creates a steady-state inventory of isotopes that would not otherwise exist on our planet.

- **The Steady State:** On a global scale, the production rate (P) of ^{14}C in the upper atmosphere is balanced by its radioactive decay (λN) and its sequestration in the oceans and biosphere.

- **The Engineering Challenge:** While these particles are a "nuisance" for spacecraft designers and chip manufacturers, they provide geochemists with a unique set of tools.

Looking Ahead: In the next lecture, we will move from the physics of the "Atmospheric Reactor" to the application of its products. We will explore the "Nuts and Bolts" of ^{14}C dating, and see how human activities—from the Industrial Revolution to the Cold War "Bomb Pulse"—have perturbed this natural clock, allowing us to track everything from the age of King Tut to the deep-sea circulation of the Atlantic Ocean.